

Blooming: Changing Laundry Habits and Opening Windows to Brighter Cities

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ABSTRACT

Laundry is a routine part of most individuals' lives and an important factor in decisions around infrastructure and property development. It's a routine that many take for granted, but in a world with increasing energy costs and environmental impact, the need to rethink the process is greater than ever. In this, tumble-dryers are associated with the greatest cost and carbon emissions in the laundry cycle while also being the most redundant. Blooming, a tumble-dryer alternative designed with cost-effectiveness, space adaptability, and sustainability at its core, is thus proposed. It is a smart-rack window installation that hangs the user's wet clothes in the draft of their home. Using sensors of the environment, the window assesses when the weather is most suitable for clothes drying and opens with the clothes rack automatically or according to the user's discretion. An accompanying app tracks the drying condition of the clothes and takes commands to close the window once finished. The app also gives feedback on the user's environmental impact, cost savings, and community progress. As such, Blooming will fit naturally in any apartment home as a utility that discourages and substitutes the costly use of drying machines while having a user journey that remains familiar and convenient.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design**; *HCI design and evaluation methods*.

KEYWORDS

behavioural change, domestic energy, laundry, sustainability

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1 INTRODUCTION

The price of electricity is currently widely regarded as one of the main contributors to the global cost of living crisis. However, electricity expenditure is also associated with the problem of climate change. This makes it a major point to tackle for both living cost reduction and sustainability practices. Thus, our research is focused on laundry practices, with a special focus on clothes drying, reflecting the major impact on the environment of the latter.

Our ideation is focused on the in-depth insights we achieved from our user survey. This explored everyday sustainable practices, habits, and challenges consistently faced in the UK. For example, our research provided awareness of users' laundry behaviors and community perceptions which made clear the distinction between users coming from different age groups and accommodation types.

Informed by our analysis, we created an adjustable automatic drying rack ergonomically fitted into the living space and controlled by a smart system with environmental sensors, the purpose of which is to allow users to protect the environment by reducing machine drying frequency while simultaneously accounting for the convenience problems of their limited domestic space.

The given research contributes to the interactive systems embedded into the "smart home" projects, ensuring not only accommodating convenience and security but also promoting sustainable practices in a non-intrusive way.

2 BACKGROUND AND MOTIVATION

Approximately 58 percent of respondents in the United Kingdom own at least one tumble dryer machine in their household, with this percentage stable for over two decades [11]. However, despite their popularity, tumble dryers are determined to be one of the largest household appliances by their energy expenditure due to their extensive electricity demand [4].

Concurrently, with increasing media awareness of climate change and the direct causal link to electrical consumption, people are becoming more informed regarding the effect their choices have on the environment [3].

Despite the extensive research regarding energy consumption minimization, there is a very limited number of technologies designed to replace the tumble dryer with a more sustainable alternative, instead vastly focusing on improving the efficiency of the existing technologies.

Thus, the motivation for this project is to develop an implementation that allows for reducing the ecological footprint of machine dryer usage, which is simultaneously convenient in execution and is aligned with the UN Sustainable Development Goal 12 in promoting sustainable consumption and production.

This report analyses and evaluates the related research in the sphere of laundry practices, consequently leading to the identification of the research gap. Furthermore, we demonstrate the process of our prototype design and evaluation, referring our ideas to the insights gained from the preliminary user survey.

Through this research, we found that space and convenience were key design requirements for new technologies hoping to change people's laundry practices. We use these requirements to guide our brainstorming and design iteration processes. As a result, we decided to develop an adjustable interactive rack to intelligibly fit into users' living spaces to minimize lifestyle disruptions and allow for the appliance's remote control through the mobile application. Finally, we conclude our research with the findings of our user testing stage and propose suggestions for future work.

3 PRIOR WORKS

While reviewing the existing research in the field of laundry practices we identified that the major body of the investigations was focused on decreasing the electricity consumption in the design and operation of tumble dryers. For example, the researcher [2] investigated the redesigning of tumble dryers by focusing on single operational aspects, such as dryer insulation, air flow mass recirculation, and waste heat utilization [8].

However, despite the significant link between economic development closely linked to mass production [9], and climate damage following the excessive ecological footprint [6], the governmental sources highlighted the dominance of domestic energy consumption (such as energy consumed by the households) over energy expenditure in the service and industry sectors [5] (Figure 1)

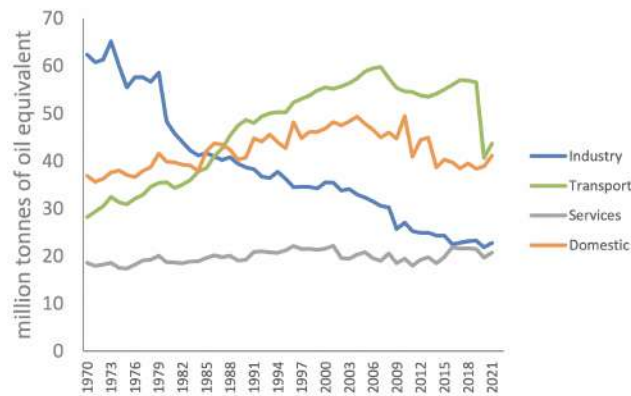


Figure 1: Energy Consumption by Sector, 1970-2021 [5]

We first reviewed the smart laundry devices designed to use together with mobile applications, such as LaunderPal [7] and Hitachi BD-120XGV [1]. The former device is designed to help users manage their laundry by providing recommendations for laundry loads

by scanning RFID, allowing laundry practices to become less time-consuming and more convenient for the user. Hitachi BD-120XGV was an AI washing machine that allowed the user to automatically select the water and detergent levels according to the laundry load and washing water properties. This serves to operate the device efficiently and ecologically without user intervention. We also inspected SmartDry [13]. SmartDry uses a smart sensor application that measures the temperature and humidity of the clothing in the tumble dryer. It switches off the system as soon as clothes are detected to be dried.

All these systems advertise sustainability as one of their preliminary contributions. However, we determined that those applications were focused on improving the existing laundry technologies, such as washing machines and tumble dryers. Therefore, they failed to address the inherent wastefulness of the energy expenditure of running the laundry machines in general, thus lacking innovative insight into the problem space.

Consequently, we examined Steigen [10], the remotely controlled automatic rack equipped with the solar heater, ionizer, and dust mite repeller. The device is advertised to reduce electricity costs by half when compared to the standard tumble dryer. Afterward, we also reviewed Blindry [12], the contemporary concept of window blinds, which are manually unfolded into the clothing rack. The given implementation was of special interest due to its ergonomic design and utilization of sunlight and air for drying. This allowed widespread utilization while being inexpensive and sustainable. The advantage of these devices is their simplicity and low energy consumption. They are also evocative of low-cost drying racks that are in wide use globally. This demonstrates that the most intuitive and efficient low-cost solutions are also low-tech. The main drawback of such a simple design was the absence of any interactive technologies to make the drying process more automated and convenient for the user.

4 USER RESEARCH

4.1 Research Method

In order to get a deeper preliminary insight into our participants' sustainability habits and struggles, we conducted a Sustainable Living questionnaire. The research sample included 32 ordinary UK residents of all ages and backgrounds; with most respondents indicating that their age is below 30 years old (21 participants). We later discarded 3 responses due to failure to comply with the consent form requirements. The questionnaire included 16 closed and 1 open question, incorporating multiple-choice, rating, ranking, and Likert scale question options.

4.2 Research Findings and Requirement Establishment

One of the major findings regarding user sustainability practices demonstrated the key differences in perceived sustainability obstacles in the different age groups. Users below the age of 30 indicated the inconvenience of sustainable practices as the major problem, followed by the complexity and lack of information issues. In contrast, older subjects expressed that additional cost was the most crucial factor limiting their sustainable behaviors, closely followed by inconvenience and time constraints. The main motivations for

Table 1: Data Table of Frequency of Dryer Usage by Domicile

Frequency	Number of responses by domicile type		
	Student accommodation	Flat	House
Always/often	6	2	1
Seldom/never	7	2	7

Table 2: Data Table of Sense of Belonging by Domicile

Sense of belonging with neighbors	Average rating from 1, disagree, to 5, agree by domicile type		
	Student accommodation	Flat	House
Familiar	1.2	1.7	3.0
Trust	1.6	1.0	2.3
Enthusiastic interactions	1.5	1.0	2.3

sustainability behaviors were defined as environmental protection and lower energy costs regardless of the respondent's age, however, the younger group was also concerned with sustainability problem space struggles. The research into the specific laundry practices indicated that the utilization of dryers correlates highly with the type of accommodation the user resides in (Table 1).

Our research consequently concluded that subjects living in private houses are less likely to use a dryer on a daily basis. Whereas student accommodation residents relied on their personal preferences. The second finding highlighted the importance of available space for sustainable laundry practices, corresponding with the significant habit distinction between the different types of domiciles.

We additionally included questions pertaining to users' attitudes toward their local community in order to probe whether solutions based on communal laundry sharing were feasible to implement. Following our findings, we ultimately abandoned this concept due to the evident lack of closeness to one's neighborhood from the questionnaire results, indicating unsubstantial levels of familiarity within user communities. Nonetheless, we decided to consider the community factor in our subsequent designs following our intention to produce a prototype with a societal background. Following our research, we found that this factor was strongly linked to the type of the respondent's accommodation. This was evidenced by the dissimilar levels of community variables across various housing arrangements.

Users residing in private houses indicated the highest familiarity, trust, and affectionate interaction (Table 2), as well as the willingness to ask for help (Table 3). Student accommodation and flat residents generally reported lower ranks for those ratings, with flat residents being less likely to ask their neighbors for help. The themes that were identified from the open question regarding the desired community support were mainly focused on information, instructions and resource availability, recycling, transportation, and social sharing systems.

Table 3: Data Table of Willingness to Ask for Help by Domicile

	Average rating from 1, unlikely, to 10, likely by domicile type		
	Student accommodation	Flat	House
Rating	4.7	3.0	7.2

4.3 Target Audience

We thus selected the profile of an average UK resident who resides in a flat accommodation type, and so generally with relatively wasteful laundry habits, as a target user. We decided on this because this profile is both prolific and has a clear design space. The other user profiles that emerged from the results conversely were less well defined, less populated, and did not correlate as strongly with the wasteful behavior that we endeavored to solve.

5 DESIGN PROCESS

5.1 Initial Ideation

Based on requirements informed by user research, we decided to focus primarily on design solutions prompting sustainable laundry practices through decreasing space scarcity and optimizing laundry runs. Initial brainstorming was done individually to avoid the potential of groupthink. Identifying overlaps between ideas, we clustered and integrated similar items together (Figure 2) for more consolidated ideas (Figure 3). Later, we utilized an affinity diagram to further organize ideas based on general themes (Figure 4). We then sorted ideas based on the user's laundry journey, from buying, wearing, washing, and drying, to storing clothes, and narrowed down to the two main steps: washing and drying, as most of our ideas, fell into these two categories. In addition, community engagement, as another theme emerging from our ideas involving collective effort, was also included in the second round of ideation.

5.2 Design Iteration

In the second round of ideation, three key ideas were introduced: 1) a smart laundry basket that detects what clothes the user wants to wash automatically and matches with others with similar needs to share laundry run together; 2) a set of smart hanging rack that detects the weather and automatically decides drying mode for users while having community engagement through the display of the physical artifact; 3) an outdoor community drying rack with an elevated mechanism enabling users to control the rack to go up and down to hang clothes (Figure 5).

To compare the ideas, we created an evaluation matrix of criteria with different weights of importance (Table 4) and evaluated the three alternatives based on defined criteria. Since we are focusing on laundry practice, as a routine of which the habit is built and strongly rooted, we considered that for the product to have maximum impact the ability to alter habit should be the most heavily weighted criterion. Though sustainability is identified as a key theme in our project, we found it difficult to compare the ideas rigorously as we were not able to quantify the exact amount of energy reduction without a concrete system design.

Table 4: Evaluation Matrix of Second Ideation

		Shared Laundry		Smart Rack		Community Rack	
Criteria	Weight	Rating	Score	Rating	Score	Rating	Score
Habit-Changing	5	2	3	1	5	3	2
Relevance to research	4	2	3	1	4	2	3
Thought Provoking	4	3	2	2	3	1	4
Scalability	3.5	1	4	2	3	3	1
Convenience	3.5	2	3.5	1	5	3	2
Community Engagement	3	1	4	2	3	2	3
Innovativeness	3	3	2	2	3	1	4
Technical Feasibility	2	1	5	2	3	3	1
Total			89.25		105		71.5

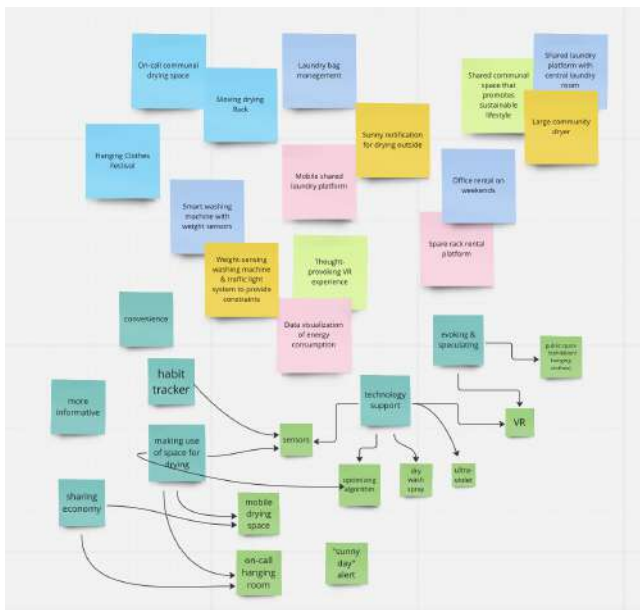


Figure 2: Initial Idea Clustering

The effect of habit-changing was analyzed based on how easy and familiar the interaction is for the user to adopt, and how accessible and convenient the design is for the user to build habits based on pre-existing routines. Since the smart rack is readily available for the user at any time, we think it is more possible for the drying habit to be changed given the possibility of high-frequency interaction. The familiarity of the drying rack would also make it easier to be adopted. In terms of relevance to research, it solves the problem of space scarcity while dodging the issues of trust that comes with the other two ideas involving a shared system. Given the above two reasons and an overall higher score, we decided to carry the smart rack idea forward.

After settling on the idea of the smart hanging rack, we did another round of brainstorming for the structure of the rack (Figure 6) with different interaction mechanisms. A 2x2 matrix (Figure 7) was then used to evaluate the compatibility of each design with the

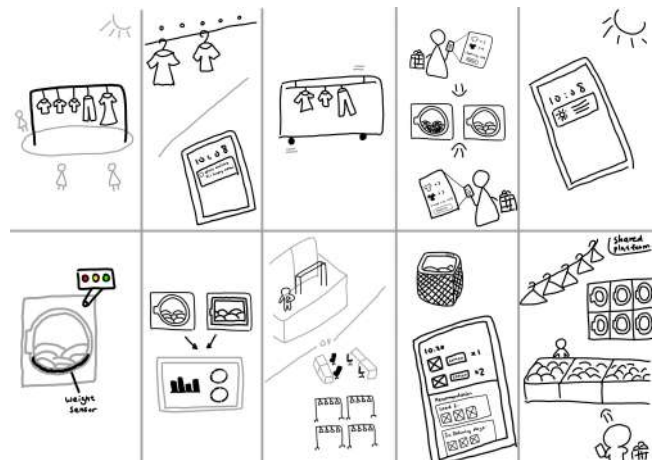


Figure 3: Initial Ideation Sketches. These sketches are (clockwise): 1) Clothes hanging festival, 2) On-call drying space 3) Bus-mounted moving drying space 4) Shared laundry platform 5) Sunny weather notification 6) Communal laundry services 7) Laundry basket management and load-size optimization 8) Spare rack/space rental 9) Data visualization of energy consumption 10) Smart washing machine with weight sensor and traffic light system prompting washing optimization.

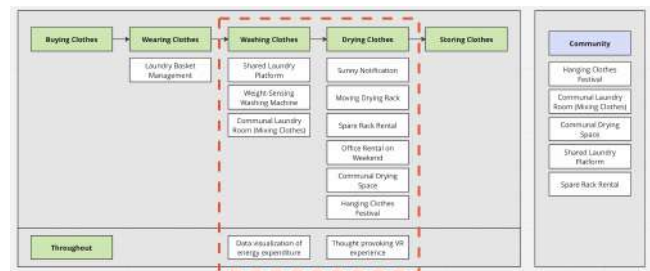


Figure 4: Affinity Diagram of Initial Ideation

project goal of reducing energy consumption, as well as the technical feasibility of each design. We proxied the reduction of energy

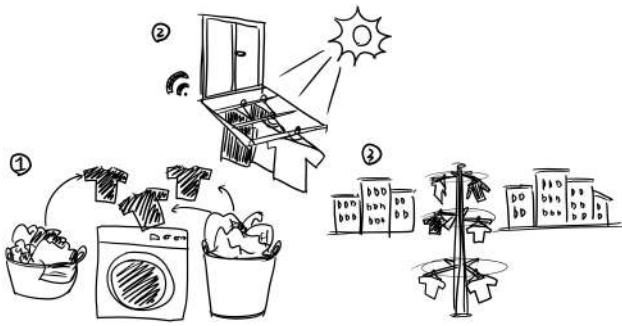


Figure 5: Sketches of Second Ideation

use with the total energy used in each design based on the hardware involved. For instance, an LED billboard display was assumed to cost more energy than basic mechanic outputs. As for feasibility, we considered the practicality of installing an additional structure on the window as well as the mechanics of the rack moving while carrying the clothes. Since only one design falls into the optimal quadrant (feasible and low energy use), we decided to adopt the structure as the physical design of the smart drying rack.

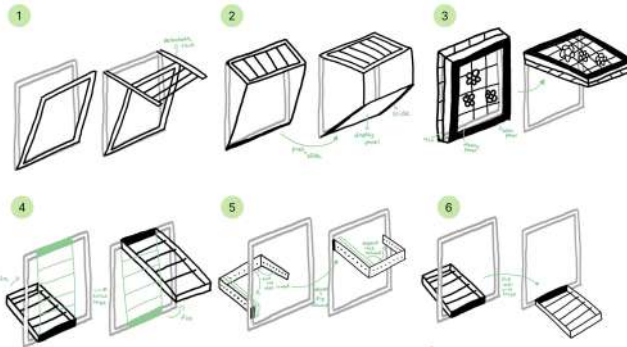


Figure 6: Brainstorming Idea Sketches of Rack Structure

For better accessibility of the rack, we incorporated a mobile application that enables remote control of the rack through an Internet connection. Later, user testing was conducted to test the design of interfaces and the overall concept with target users.

Results from the study showed that there is a hierarchy of needs when it comes to interacting with the system. As the main goal is to hang clothes, participants elicited a greater need to control via a physical button instead of the mobile application. In addition, participants felt that it is less important to see the drying progress on their mobile since the system is automatic in nature.

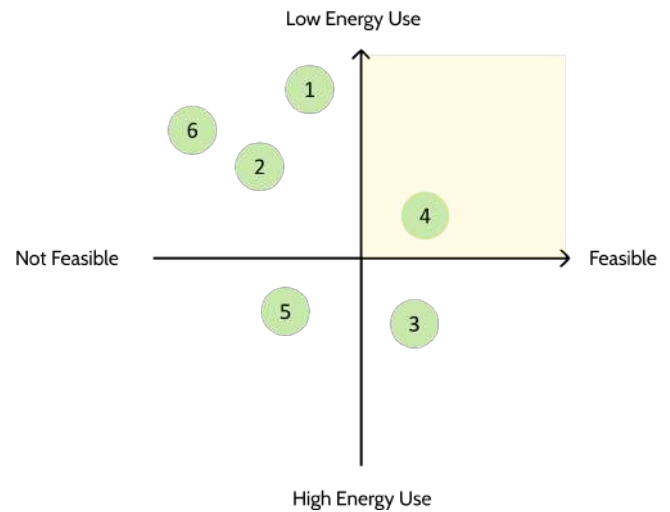


Figure 7: 2x2 Matrix of Structure Design

6 FINAL DESIGN

6.1 Prototype

Our final design is Blooming, a smart air-drying system that replaces traditional tumble dryers, reducing electricity consumption and financial cost. It is constituted of two parts: a physical smart rack installed on the window and a mobile application (Figure 8). Blooming makes the best use of the weather condition for outdoor drying through real-time sensing. Additionally, it keeps track of the drying progress. Hinge mechanisms attaching the rack to the window allow the rack to automatically rotate from a position inside the room, to be loaded with clothes, to a position outside the window, to dry them. If the clothes are detected as dry, it will automatically send them back to the room for the user to collect. With both physical buttons installed near the rack and the app, the user can access the system whenever and wherever they want, from standing beside the rack with a basket of wet clothes to being remotely in their offices.



Figure 8: Prototype: Model of the Smart Rack and the Accompanied Mobile Application.

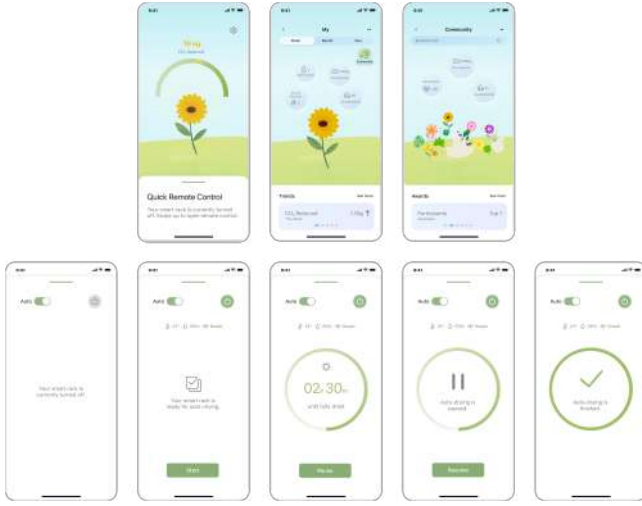


Figure 9: User Interface

In addition to automatic drying, Blooming also keeps track of users' monthly air-drying history (Figure 9). Visualization of users' efforts to reduce energy will be reflected on both the physical rack and the app, to give them a sense of accomplishment. On the physical rack, there are mechanical "flowers" that bloom in response to users' progress: the more CO₂ emission they reduce, the more flourishing the flowers will be. Similar imagery is also shown on the app, with a customized flower growing as the user saves more energy. A community section is also included where users can see the flower representing their neighbors' progress within a virtual community garden.

The blooming of the flower was chosen as the main conceptual theme. This analogy was drawn as the drying system being activated by a promising period of sunlight reminded us of flowers blooming as a response to the sun. There is also a metaphorical resemblance between the expansive gestures of the operating rack and the opening petals of a blooming flower.

6.2 State Transition Design and Modeling

Based on the user journey, we mapped all the possible interaction touchpoints into a state-transition diagram (Figure 10). The vertical workflow on the left illustrates the two main functions of the mobile app: tracking sustainable actions and interacting with the rack.

To make it easier to learn and use, we simplified the home page by leaving only a personalized flower and a slide-up page. The flower pattern serves as a visual representation of the user's energy-saving achievement. This is a measure of the user's savings over using a traditional drying machine. The slide-up page shows the current stage of the drying process and leads the user to a new page to control the rack by switching states or modes. By clicking the flower, the user will enter the "individual tracking" page to view more details about their sustainable drying history. A "community achievement" page is also accessible for users to see the flowers representing their neighbors' progress as well as the neighborhood's total carbon savings as a whole. As for users' interaction with the rack, we used the state-transition diagram in the lower right of (Figure 10)

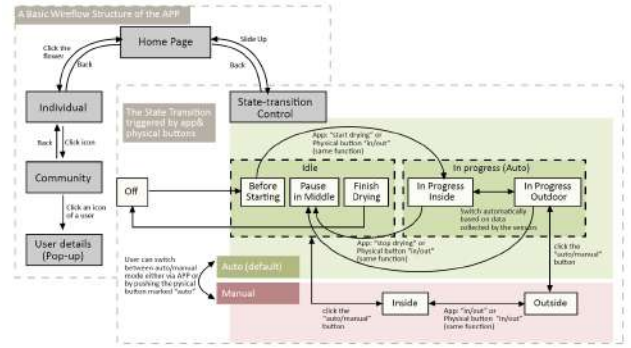


Figure 10: Interaction Logic and State-Transition Design

to ensure all the circumstances are covered. In addition to the app, we included three physical buttons as a "backup mechanism" in case the Internet-of-Things system crashes or users are simply not using the mobile app accompaniment. For the same reason, we also decided to add a "manual mode" in addition to the default "auto mode". This is to ensure that users can control the racks even if something goes wrong, such as if the rack becomes faulty and locks into position and needs to be forced open.

With that in mind, we set up four states for the default auto mode:

- Off: The power is off, and the rack is contracted.
- Idle: The power is on, yet no automatic operation is taking place. This state consists of three nuances: before drying; pause drying; finish drying. This is to ensure the rack does not automatically move until the user finishes hanging up (or taking off) all the clothes. It will also prevent the rack from moving when activated while a user is in the progress of hanging clothing on it.
- In progress (auto mode): The system moves inwards/outwards automatically based on the weather and the drying condition of the clothes as detected by sensors.

While "auto mode" is the default, users can also choose to control the position of the rack manually. In this case, the system stops auto-sensing and decision-making.

6.3 User Journey Illustration

The storyboard (Figure 11) demonstrates a typical usage flow where a flat resident (user A) uses the Blooming rack to dry his clothes. It begins when user A receives a message acknowledging that the weather will be suitable for air-drying clothes. He then realizes how he can make good use of the sunny weather for laundry. After washing the clothes, he turns the systems on, hangs the clothes on the rack, and presses "start" to let the system do its work.

For the next few hours, the rack stays outside until the clothes are detected to be dry and are automatically sent indoors, at which point user A receives a notification. Every time user A finishes drying his clothes with the rack, the system will calculate the amount of CO₂ emission saved using the weight of the clothes detected by integrated sensors. This achievement is also recorded and visualized as the growth of the flower on the home page of the app. After

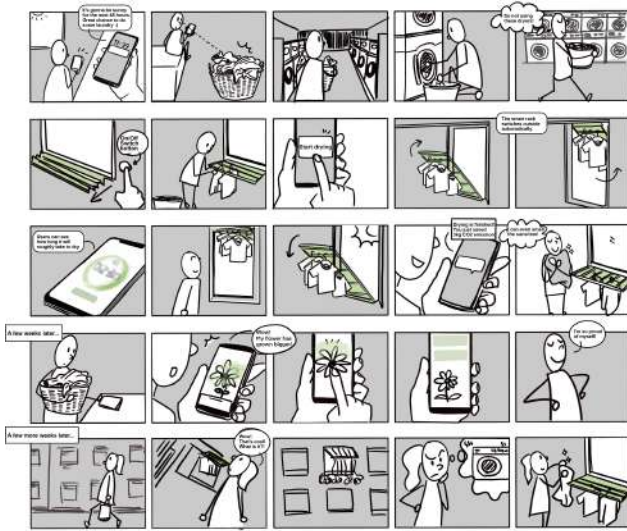


Figure 11: Storyboard of the User Journey

using the system for a few weeks, user A finds the flower significantly bigger and is proud to make these frequent and habitual, yet quantifiable, contributions to environmental protection.

A few days later, another resident of the same neighborhood (User B) passes by and sees the mechanical flowers on user A's window. This physical decoration on the building facade is so impressive that User B gets motivated to give this new method a try and starts to reflect on her current drying choice. She ends up pleased with how much CO₂ she can save with such a simple behavioral change. Gradually, the building is revitalized by flowers blooming on multiple tenants' windows.

7 DISCUSSION

Blooming provides users with a low-effort, but high-impact, solution for clothes drying that encourages eco-friendly lifestyles and has implications on laundry practices as a whole. Throughout the entire design and research process specific attention was given to keeping Blooming and the associated app simple and intuitive to use. As such, Blooming will fit naturally in any apartment home as a utility that discourages and substitutes the costly use of drying machines while having a user journey that remains familiar and convenient.

We note the boundary to installation of Blooming in homes where fixing it into a wall would require cooperation from the community, property developers, and the user. It would be logistically more intensive than installing a tumble dryer. However, in the long run, Blooming can benefit both property developers and the occupants financially and environmentally. Whether the initial installation costs are covered by the landlord or occupant, Blooming costs considerably less than traditional tumble dryers and pays for itself through the reduction in bills. Nonetheless, where different parts of a building get exposed to more or less sunshine or better or worse weather, a greater environmental assessment is likely to impact the interest in, and efficiency of the use of, Blooming.

From our background research and initial survey, our problem and design spaces were clear. People living in urban environments seeking low-cost accommodation are encouraged to foot the financial and environmental cost of wasteful laundry practices due to the lack of space that they end up with due to their choice of home. Furthermore, we found that generally, the most intuitive and efficient options were also the most low-tech. This challenged us to create a solution that users would feel comfortable using habitually, and that could foster the uptake of eco-friendly attitudes and lifestyles in the community, with a technology-centric approach.

In the prototyping and user-testing phase, we also experimented with the use of an associated laundry-management app. This app gives feedback on the drying condition of the user's clothes, and the financial and ecological impact of the use of Blooming, and allows the user to operate the device remotely. This helped us ensure that the technology was meaningful and relevant to the user. For example, where we found that many users would prefer using a manual interface for Blooming over a digital one we added the manual button controls to the window. The app also gives the user a sense that something as mundane as their laundry practices had bigger-picture implications and so could encourage an eco-friendlier lifestyle and community engagement.

We would want to continue to observe user behavior with our technologies and identify pain points that could emerge from continued use. These may be to do with the maintenance of the product or the rate of uptake of the technology. Further work could investigate the attitudes of the different logistical stakeholders in the device. As such, uptake is both an issue of whether it truly changes user habits and whether stakeholders are interested in installing it in their buildings. We are also interested in developing community-oriented features of the app, such as sharing one's environmental progress with their local neighborhood and collaborating to improve the community's carbon footprint. This would require further study into the user's privacy and social implications otherwise. For example, as long as the use of Blooming is visible to outside observers, users may be conscious of the impression their use of it makes to others.

We could also consider how more home utilities could be interfaced with our app and implement more integration with the IoT. Getting greater uptake of Smart-Home systems, in general, may only be possible with a principle for low-cost, high-impact, minimalist solutions, as we used in the development of our Smart-Rack, at the core of its design. Social engagement is also an entirely non-superficial factor and gives users some purpose and motivation.

8 CONCLUSION

In this article, we explored the solutions to achieve UN sustainable development goal 12: responsible consumption and production. In particular, we explored possibilities in uptaking smart domestic utilities to pursue the goal in a low-cost, high-impact, and minimalist way. In response to the literature review and user research, we proposed Blooming, a tumble-dryer alternative designed with cost-effectiveness, space adaptability, and sustainability at its core. Blooming helps researchers and designers to reflect on how behavior change intervention influences people to live in a more sustainable way and explore more effective solutions to achieve

the sustainable goal. It will inspire more design practices to solve energy consumption problems.

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